

Coding Round Practice Set

Level 2 – One Notch Harder

20 Scenario-Based Questions, One Level Above the Original Set

(TCS NQT · Infosys · Wipro Elite NLTH · Cognizant GenC · Accenture · Capgemini style)

How to use this set: Same format as before — problem statement, input/output format, constraints, and sample input-output. These build on the same topics but add a twist (edge cases, larger constraints, combined concepts, or an extra condition) so they sit one level above the first set.

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1	Valid Palindrome with Noise	Strings	Medium
2	Group the Word Tiles	Strings / Hashing	Medium
3	Third-Highest Sales – SuperMart Regional Report	Arrays	Medium
4	Top-3 Reported Issues – IT Helpdesk Analytics	Hashing	Medium
5	Prime Range Report – Gate Access Audit	Math	Medium
6	Distinct Arrangements – Workshop Scheduler v2	Math / Recursion	Medium
7	Bacterial Growth with Mutation – BioLab Research	Recursion / DP	Medium
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10	Multi-Key Leaderboard – GameZone Arena	Sorting	Medium
11	Employee ID Lookup in a Rotated Log – Zenith Corp	Searching	Medium-Hard
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Questions

Q1. Valid Palindrome with Noise

Topic: Strings | *Difficulty:* Medium

BrightPack Logistics now receives product codes scraped from scanned labels, which sometimes pick up stray punctuation, spaces, or OCR artifacts. A code should still be considered a palindrome if it reads the same forwards and backwards after removing every non-alphanumeric character and ignoring case.

Input Format:

A single line containing a string S of length 1 to 1000 (may contain letters, digits, spaces, and punctuation).

Output Format:

Print "PALINDROME" if S is a palindrome under the above rule, otherwise "NOT A PALINDROME".

Constraints:

$1 \leq |S| \leq 1000$

Sample Input	Sample Output
A man, a plan, a canal: Panama	PALINDROME
Race a car	NOT A PALINDROME

Q2. Group the Word Tiles

Topic: Strings / Hashing | *Difficulty:* Medium

WordWhiz wants to extend its "Twin Words" badge into a full Anagram Grouping feature: given a list of words, group every word with all of its anagrams, and display each group on one line in the order the group's first member first appeared. Words within a group should also preserve their original relative order.

Input Format:

First line: integer n . Next n lines: one word each (lowercase letters only).

Output Format:

Print each group's words space-separated on its own line, groups ordered by first appearance.

Constraints:

$1 \leq n \leq 10^4$, $1 \leq \text{length of each word} \leq 50$

Sample Input	Sample Output
6 eat tea tan ate nat bat	eat tea ate tan nat bat

Q3. Third-Highest Sales – SuperMart Regional Report

Topic: Arrays | *Difficulty:* Medium

The regional manager now wants to reward the top three performing branches. Find the third-highest distinct sales value from the list. If fewer than 3 distinct values exist, print “NOT ENOUGH DISTINCT VALUES”.

Input Format:

First line: integer n. Second line: n space-separated integers.

Output Format:

The third-highest distinct value, or the not-enough message.

Constraints:

$1 \leq n \leq 10^5$, $0 \leq \text{sales}[i] \leq 10^9$

Sample Input	Sample Output
6 45000 67000 67000 52000 39000 70000	52000
2 10 10	NOT ENOUGH DISTINCT VALUES

Q4. Top-3 Reported Issues – IT Helpdesk Analytics

Topic: Hashing | *Difficulty:* Medium

Nova Solutions now wants the top 3 most frequently reported issue categories, in descending order of frequency. If categories tie in frequency, list the smaller category code first. If fewer than 3 distinct categories exist, list only what's available.

Input Format:

First line: integer n. Second line: n space-separated integers.

Output Format:

Up to 3 category codes, space-separated, most frequent first.

Constraints:

$1 \leq n \leq 10^5$

Sample Input	Sample Output
10 101 102 101 103 102 101 104 103 103 105	101 103 102

Q5. Prime Range Report – Gate Access Audit

Topic: Math | *Difficulty:* Medium

Security wants a weekly audit tool: given a range [L, R], report how many valid (prime) access codes exist in that range, and list the largest one. Efficiency matters — a naive per-number check must scale to the full range.

Input Format:

Two space-separated integers L and R.

Output Format:

Print "Count: C, Largest: X". If no prime exists in the range, print "Count: 0, Largest: NONE".

Constraints:

$$1 \leq L \leq R \leq 10^6$$

Sample Input	Sample Output
10 30	Count: 6, Largest: 29

Q6. Distinct Arrangements – Workshop Scheduler v2

Topic: Math / Recursion | *Difficulty:* Medium

Some of the N tasks are actually identical (same task type repeated). Given the task list with possible duplicates, compute the number of distinct orderings — i.e., $N!$ divided by the factorial of each duplicate group's size. Since the answer can be huge, print it modulo 1,000,000,007.

Input Format:

First line: integer N. Second line: N space-separated task-type integers.

Output Format:

The distinct arrangement count mod $1e9+7$.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input	Sample Output
4 1 1 2 3	12

Q7. Bacterial Growth with Mutation – BioLab Research

Topic: Recursion / DP | *Difficulty:* Medium

The lab now models a colony where the population at hour i is $f(i) = f(i-1) + f(i-2) + f(i-3)$ (a tribonacci-style rule), starting from given seed values for hours 0, 1, and 2. Since N can be large, compute $f(N)$ mod 1,000,000,007 using an efficient (non-exponential) method.

Input Format:

First line: three space-separated integers, the seed values $f(0)$, $f(1)$, $f(2)$. Second line: integer N.

Output Format:

The value of $f(N)$ mod $1e9+7$.

Constraints: $0 \leq \text{seeds} \leq 10^9, 0 \leq N \leq 10^6$

Sample Input	Sample Output
0 1 1 10	149

Q8. Armstrong Range Certification – QuantumTech Electronics

Topic: Math | Difficulty: Medium

QA now wants to certify an entire batch at once: given a range [L, R] of serial numbers, print all Armstrong numbers in that range (space-separated, ascending order), and their count. If none exist, print "NONE FOUND".

Input Format:

Two space-separated integers L and R.

Output Format:

First line: the Armstrong numbers found, space-separated (or "NONE FOUND"). Second line: "Count: C".

Constraints: $1 \leq L \leq R \leq 10^7$

Sample Input	Sample Output
100 500	153 370 371 407 Count: 4

Q9. In-Place 90° Rotation – Image Preview

Topic: 2D Arrays | Difficulty: Medium

PixelCraft now needs the actual 90-degree clockwise rotation (not just the transpose), and it must be done in place (without allocating a second full matrix) for a square N×N pixel matrix, to keep memory usage low on mobile devices.

Input Format:

First line: integer N. Next N lines: N space-separated integers each.

Output Format:

The matrix after an in-place 90° clockwise rotation, N rows of N space-separated integers.

Constraints: $1 \leq N \leq 500$

Sample Input	Sample Output
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3	7 4 1
1 2 3	8 5 2
4 5 6	9 6 3
7 8 9	

Q10. Multi-Key Leaderboard – GameZone Arena

Topic: Sorting | Difficulty: Medium

The leaderboard now needs to break ties fairly: sort players primarily by score (descending); if scores tie, sort by number of matches played (ascending — fewer matches played is more impressive); if still tied, sort alphabetically by player name.

Input Format:

First line: integer n . Next n lines: name score matches (space-separated; name has no spaces).

Output Format:

n lines, each name score matches, sorted as described.

Constraints:

$1 \leq n \leq 10^5$

Sample Input	Sample Output
4 Alex 80 10 Ravi 80 6 Zoe 95 8 Amit 80 6	Zoe 95 8 Amit 80 6 Ravi 80 6 Alex 80 10

Q11. Employee ID Lookup in a Rotated Log – Zenith Corp

Topic: Searching | Difficulty: Medium-Hard

Due to a backup restore, the sorted employee ID log got circularly rotated at an unknown pivot (it's still made of a sorted sequence, just rotated). Find the 1-indexed position of a target ID using an algorithm no slower than $O(\log n)$ — a plain linear scan will not be accepted.

Input Format:

First line: integer n . Second line: n space-separated integers (a rotated, strictly increasing sequence, all distinct). Third line: target ID.

Output Format:

"Found at position X " (1-indexed, position in the given rotated array) or "NOT FOUND".

Constraints:

$1 \leq n \leq 10^6$

Sample Input	Sample Output
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7 110 118 125 130 101 102 105 101	Found at position 5
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Q12. Longest Palindromic Review Snippet

Topic: Strings | **Difficulty:** Medium-Hard

The content-quality team wants to auto-highlight the longest palindromic substring within a customer review (case-insensitive, letters only — ignore digits, spaces, and punctuation entirely when scanning, but report the original substring text from the cleaned letters-only version). If multiple longest palindromes exist, report the first (leftmost) one.

Input Format:

A single line of review text (up to 5000 characters).

Output Format:

The longest palindromic substring (from the cleaned, letters-only, lowercase text).

Constraints:

$1 \leq \text{length} \leq 5000$

Sample Input	Sample Output
Was it a car or a cat I saw	wasitacaroracatisaw

Q13. Minimum Fixes for Syntax Checker

Topic: Stack | **Difficulty:** Hard

The browser-based editor now wants to be more helpful: instead of just saying a snippet is unbalanced, it should report the minimum number of single-character insertions needed to make the bracket sequence `(){}[]` balanced.

Input Format:

A single line containing only the characters `(){}[]`.

Output Format:

Print a single integer — the minimum insertions required (0 if already balanced).

Constraints:

$1 \leq \text{length} \leq 10^5$

Sample Input	Sample Output
{{()}}	4
{{()}}	0

Q14. Selective Reverse Mode – PingTalk Chat App

Topic: Strings | Difficulty: Medium

An upgraded “Reverse Mode” reverses the order of words but only for words made entirely of alphabetic characters; words containing digits, punctuation, or symbols must stay in their original position (as if they were fixed anchors), while the alphabetic words rotate among the remaining slots in reverse order.

Input Format:

A single line — a sentence with words separated by single spaces.

Output Format:

The sentence after selective word reversal.

Constraints:

$1 \leq \text{number of words} \leq 1000$

Sample Input	Sample Output
Hello 123 how are you2 doing well	well 123 doing are you2 how Hello

Q15. Newsletter Cleanup with Frequency Cap

Topic: Arrays / Hashing | Difficulty: Medium

BrightAds now allows an email to appear up to K times (not just once) in the cleaned list — anything beyond the Kth occurrence should be dropped, while preserving original order.

Input Format:

First line: two integers n and K. Next n lines: one email each.

Output Format:

The cleaned list (one email per line, original order, each capped at K occurrences), followed by "Final count: X".

Constraints:

$1 \leq n \leq 10^5$, $1 \leq K \leq n$

Sample Input	Sample Output
7 2 a@x.com b@x.com a@x.com a@x.com c@x.com b@x.com b@x.com	a@x.com b@x.com a@x.com c@x.com b@x.com Final count: 5

Q16. Multi-Room Tile Sizing – DecorNest Interiors

Topic: Math | **Difficulty:** Medium

DecorNest is now quoting an entire apartment with multiple rectangular rooms and wants one single square tile size that works without cutting in every room. Given the dimensions of N rooms, find the largest square tile side length (the GCD across all room dimensions) and the LCM of all dimensions (for bulk batch planning).

Input Format:

First line: integer N (number of rooms). Next N lines: two integers L and W (each room's length and width).

Output Format:

Print "GCD: g, LCM: l" where g and l are computed across all 2N dimension values. If the LCM would overflow a 64-bit signed integer, print "LCM: TOO LARGE" instead of the value.

Constraints:

$1 \leq N \leq 1000$, $1 \leq L, W \leq 10^6$

Sample Input	Sample Output
2 18 24 12 30	GCD: 6, LCM: 360

Q17. Diamond Seat Preview – EventPro

Topic: Pattern Printing | **Difficulty:** Medium

EventPro's design team wants a diamond-shaped number pattern for a special circular auditorium layout: it should expand from 1 seat up to N seats per row and then symmetrically contract back down to 1, centered and padded with spaces so the diamond shape is visually correct.

Input Format:

A single integer N (the widest row's seat count).

Output Format:

The diamond pattern, one row per line, numbers space-separated and centered with leading spaces.

Constraints:

$1 \leq N \leq 15$

Sample Input	Sample Output
3	1 1 2 1 2 3 1 2 1

Note: Leading spaces should align each row so the numbers form a diamond; exact spacing is validated by the judge, not by strict character count.

Q18. Check Digit with Persistence Count – SecureBank

Topic: Math | Difficulty: Medium

SecureBank also wants to know the additive persistence — the number of summing steps it took to reach the digital root — alongside the digital root itself, for internal debugging of the check-digit algorithm.

Input Format:

A single integer N.

Output Format:

Print "Digital Root: R, Persistence: P".

Constraints:

$1 \leq N \leq 10^{18}$

Sample Input	Sample Output
9875	Digital Root: 2, Persistence: 2

Q19. Closest Expense Pair – SplitEasy App

Topic: Arrays / Hashing | Difficulty: Hard

SplitEasy now handles the realistic case where no exact pair may sum to the target. Find the pair of expenses whose sum is closest to the target (absolute difference minimized). If multiple pairs tie for closest, print the one that appears first when scanning left to right by the first element's index, then the second element's index.

Input Format:

First line: integer n. Second line: n space-separated integers. Third line: integer target.

Output Format:

The two expense values (in the order they appear in the list) forming the closest-sum pair, and their absolute difference from the target, in the format "X Y (diff: D)".

Constraints:

$2 \leq n \leq 10^5$, $1 \leq \text{expense}$, $\text{target} \leq 10^7$

Sample Input	Sample Output
5 40 25 15 60 35 77	40 35 (diff: 2)

Q20. Smart Parking with VIP Priority – CityMall

Topic: Simulation | Difficulty: Hard

CityMall has added a VIP tier: slots 1 to K are reserved exclusively for VIP cars, while slots K+1 to N are for regular cars. A PARK VIP command must allot the lowest free slot in [1, K] (or print "FULL" if none free); a PARK REGULAR command must allot the lowest free slot in [K+1, N] (or "FULL" if none free). LEAVE x frees

slot x regardless of tier. Process the queries efficiently (a linear scan per query over up to 10^4 slots for up to 10^4 queries should still run comfortably within limits, but avoid anything worse than that).

Input Format:

First line: three integers N , K , Q (total slots, VIP slot count, number of queries). Next Q lines: PARK VIP, PARK REGULAR, or LEAVE x .

Output Format:

For every PARK command, print the allotted slot number or "FULL", in order.

Constraints:

$$1 \leq K < N \leq 10^4, 1 \leq Q \leq 10^4$$

Sample Input	Sample Output
5 2 6	1
PARK VIP	2
PARK VIP	FULL
PARK VIP	3
PARK REGULAR	1
LEAVE 1	
PARK VIP	

Tip: These are designed for 20–30 minutes each — most add either an efficiency constraint (no brute force), an extra edge case, or a combination of two concepts from the original set.